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SMART CAMPUS AI

Problem Analysis

Universities receive a large number of student queries every day regarding course registration, class timetables, examination schedules, attendance, fee details, and campus services. Handling these queries manually is time-consuming and increases the workload of administrative staff. Students may also experience delays in getting information, especially outside office hours.

An AI-based Student Support Assistant solves this problem by providing instant, accurate, and 24×7 responses. The chatbot uses Natural Language Processing (NLP) to understand student questions, identify their intent, and return the appropriate information. This improves communication, reduces waiting time, and enhances the overall student experience.

Agent Design

Agent Name: SmartCampus AI – Student Support Assistant

Objective

- Provide instant responses to common student queries.
- Reduce the workload of administrative staff.
- Offer support 24×7.

Users

- Students
- Faculty members
- Administrative staff

Environment

- University website
- Student portal
- Mobile application

Inputs

- Student text queries such as:
- "How can I register?"
- "Show my timetable"
- "When are exams?"

Outputs

- Course registration information
- Timetable
- Examination schedule
- Attendance details
- Fee details
- Campus services information

Performance Measures

- High response accuracy
- Fast response time
- Easy interaction
- High user satisfaction
- Correct intent recognition

2.Dialogflow Agent Development

The chatbot was developed using Google Dialogflow ES. Dialogflow uses NLP to identify the user's intent and generate suitable responses.

Intents Created

1. Default Welcome Intent
2. Course Registration
3. Timetable
4. Attendance
5. Examination Schedule
6. Fee Details
7. Campus Services
8. Default Fallback Intent

Training Phrases

Each intent contains multiple training phrases to recognize different ways of asking the same question.

Example (Course Registration):

- How can I register?
- Register my course
- Course registration
- Registration process
- Register for classes

Responses

- Each intent provides a predefined response.
- **Example**
- **User:** How can I register?
- **Bot:** Course registration starts on 10 August. Login to the Student Portal, select your semester, choose your subjects, submit the registration, and download the acknowledgement receipt.

Working of the Agent

- User enters a question.
- Dialogflow compares it with the training phrases.
- The matching intent is identified.
- The chatbot returns the predefined response.
- If no intent matches, the Default Fallback Intent is triggered.
- Include screenshots of:
- Intents list
- Welcome Intent
- Course Registration test
- Timetable test
- Attendance test
- Examination Schedule test
- Fee Details test
- Campus Services test
- Default Fallback test

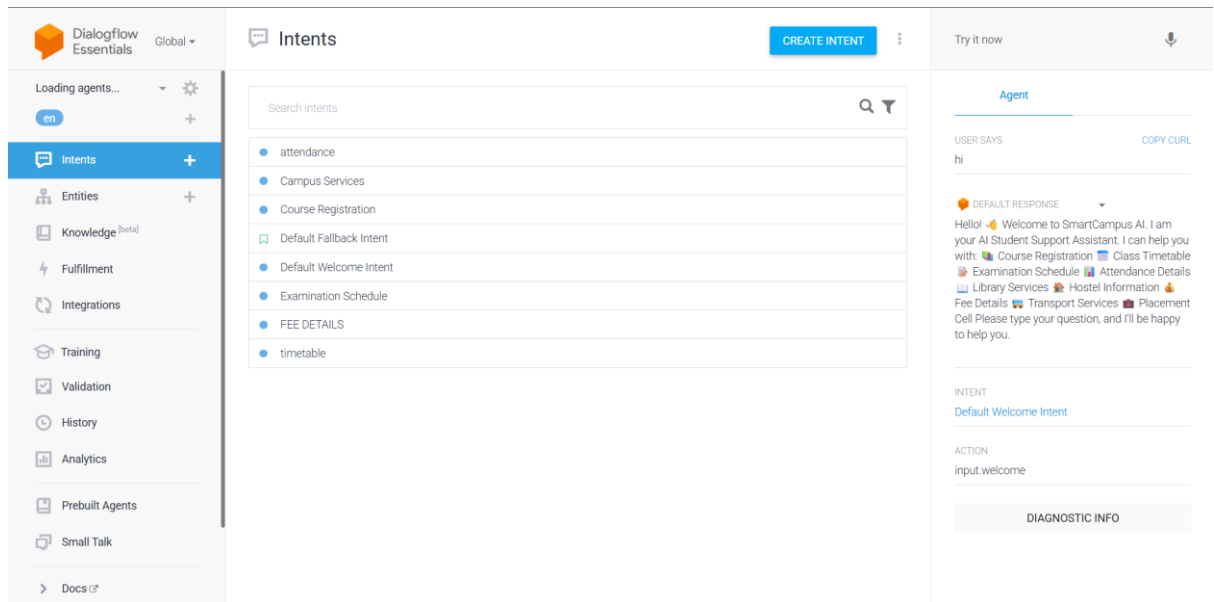


Fig1:Smart campus AI agent

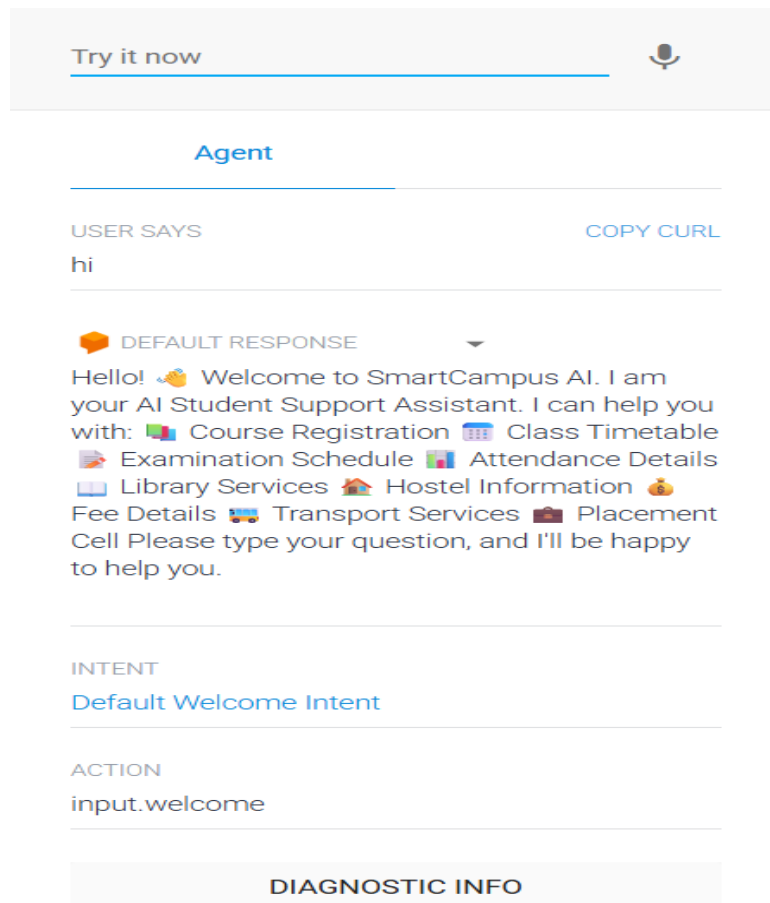


Fig2: Testing the welcome intent

Default Fallback Intent

SAVE

DEFAULT

+

Text Response

1

I didn't get that. Can you say it again?

2

I missed what you said. What was that?

3

Sorry, could you say that again?

4

Sorry, can you say that again?

5

Can you say that again?

6

Sorry, I didn't get that. Can you rephrase?

7

Sorry, what was that?

8

One more time?

9

What was that?

10

Say that one more time?

11

I didn't get that. Can you repeat?

12

I missed that, say that again?

Fig 3: Default fallback intent

3.Problem Formulation and Decision Flow:

Initial State

The chatbot is waiting for a user query.

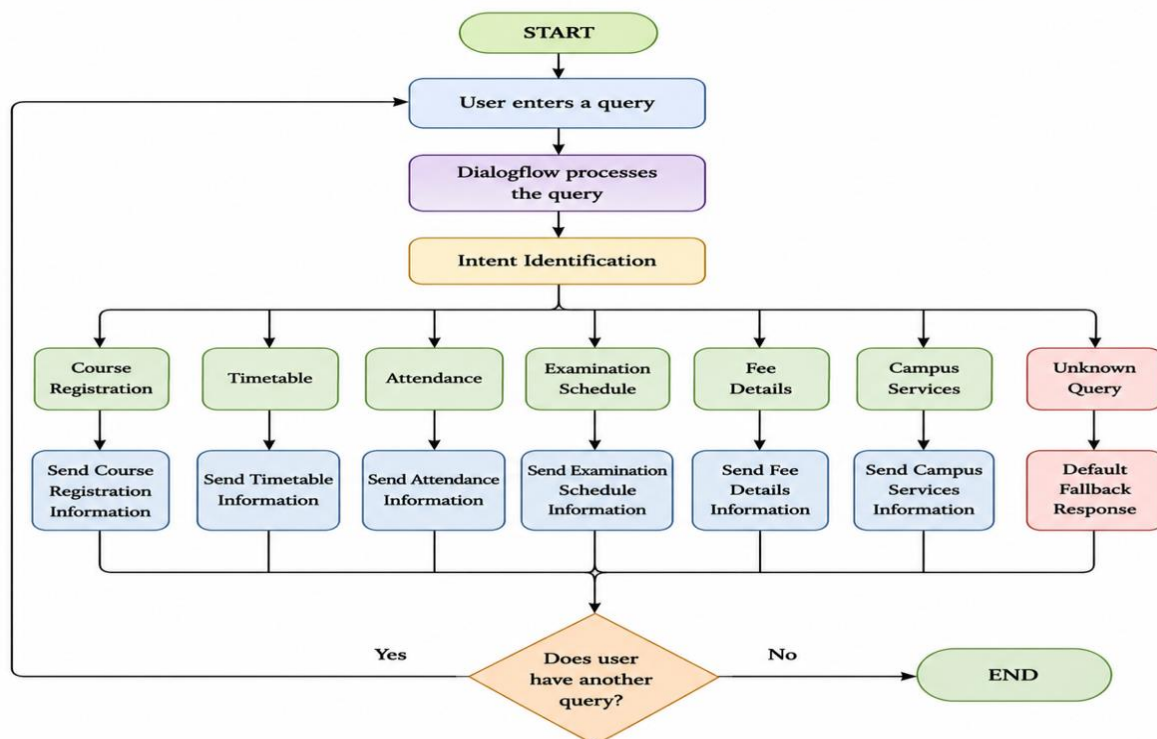
Goal State

The user receives the correct information.

Possible Actions

- Ask about course registration
- Ask about timetable
- Ask about attendance
- Ask about examinations
- Ask about fees
- Ask about campus services

CONVERSATION FLOW DIAGRAM (STATE TRANSITION)



Goal Completion

The conversation ends after the chatbot provides the required information or the user exits.

4.Search Strategy Application:

The chatbot uses Breadth-First Search (BFS) as the search strategy for dialogue navigation.

Why BFS?

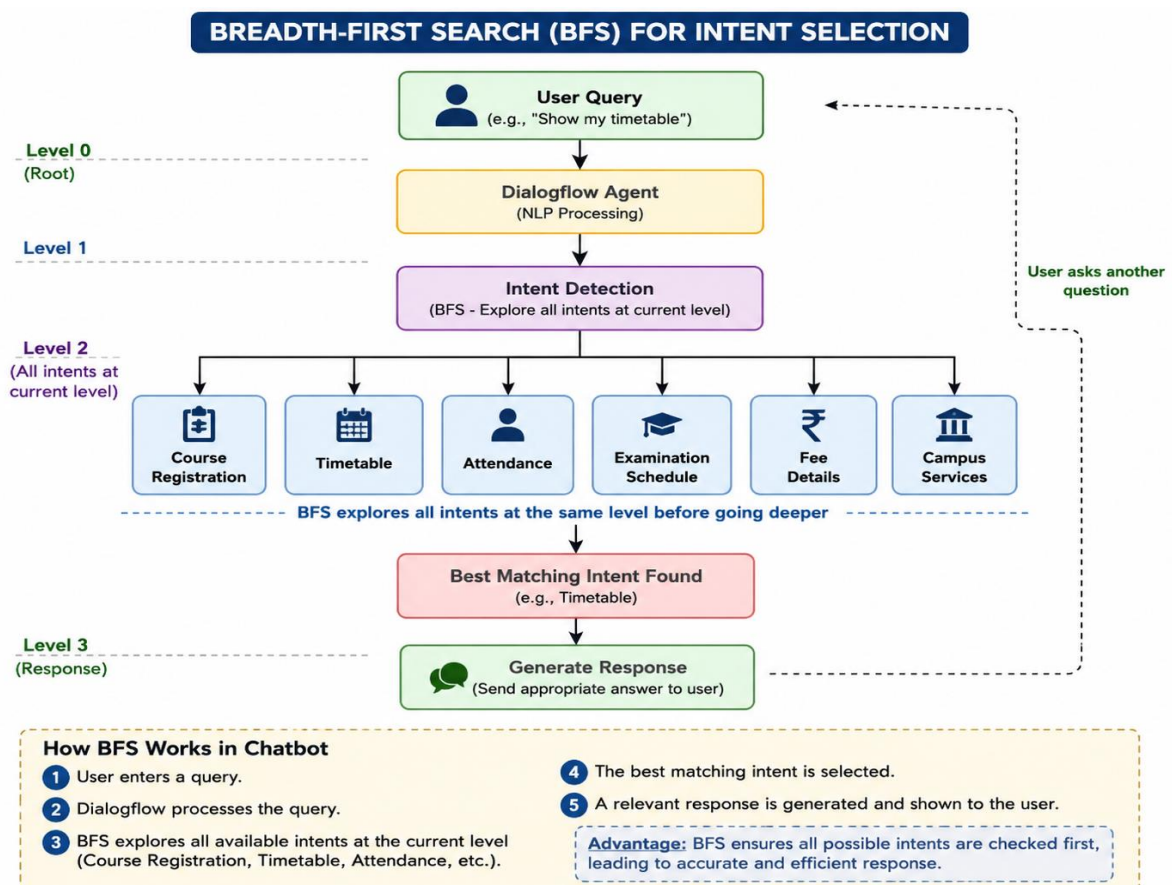
Breadth-First Search explores all possible intents at the current level before moving to deeper levels. This helps Dialogflow identify the most appropriate intent for a user's query efficiently.

Advantages of BFS in Chatbots

- Fast intent matching
- Better dialogue navigation
- Avoids missing relevant intents
- Improves response accuracy
- Simple and effective for structured conversations

Justification

In this chatbot, when a student enters a query, Dialogflow compares it against the available intents (Course Registration, Timetable, Attendance, Examination Schedule, Fee Details, Campus Services, etc.). BFS provides a systematic way to explore possible conversation paths and select the best matching response, resulting in quick and accurate assistance.



5. Testing and Evaluation:

Test Query	Expected Result	Status
Hi	Welcome	Passed
Register my course	Registration	Passed
Show my timetable	Timetable	Passed
Show my attendance	Attendance	Passed
When are exams?	Exam Schedule	Passed
Show fee details	Fee Details	Passed
Campus services	Campus Services	Passed
abcdef123	Default Fallback	Passed

CONCLUSION

The SmartCampus AI Student Support Assistant was successfully designed and implemented using Google Dialogflow ES to provide instant and intelligent support for students. The chatbot is capable of understanding natural language queries and responding accurately to common student requests related to course registration, class timetable, attendance details, examination schedule, fee details, and campus services. By using Natural Language Processing (NLP), the system identifies the user's intent and provides the most appropriate response within a few seconds, making the interaction simple, efficient, and user-friendly.

During the implementation, multiple intents were created with suitable training phrases and responses to handle different types of student queries. The chatbot was thoroughly tested with various inputs, and the results showed that it correctly recognized the user's intent and generated the expected response. Unknown or unsupported queries were handled through the Default Fallback Intent, ensuring that the conversation remained meaningful even when the input was not recognized. The Breadth-First Search (BFS) concept was used to explain how the chatbot systematically explores available intents and selects the most relevant response for the user's query.

This project demonstrates how AI-powered conversational agents can improve communication between students and educational institutions by providing 24×7 assistance, reducing the workload of administrative staff, and minimizing response time. The chatbot enhances the student experience by offering quick access to essential academic information without requiring manual intervention. In the future, the system can be further improved by integrating it with the college database to provide personalized information such as individual attendance, examination results, and fee status. Additional features such as multilingual support, voice interaction, and mobile application integration can also be added to make the chatbot more intelligent, interactive, and accessible to all students.

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